UNBC UNIVERSITY OF NORTHERN BRITISH COLUMBIA

ECON 204 – Contemporary Economic Issues

Final Exam

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Student Name: ______________________________

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Instructions:

- This exam contains **75 multiple choice questions and one essay question**.
- You have **120 minutes** to complete the exam.
- Choose the **ONE best answer** for each question.
- Fill in the bubble completely using a dark pencil.
- No communication devices are permitted.

Solutions and Answer Key

1. **C**. Inflation:
$$\frac{159.8 - 146.2}{146.2} \times 100 = \frac{13.6}{146.2} \times 100 \approx 9.3\%.$$

2. **D**. Nominal wage growth:
$$\frac{64,200 - 61,500}{61,500} \times 100 \approx 4.39\%.$$
Real wage growth:
$$4.39\% - 9.3\% \approx -4.9\%.$$
Closest answer: -5.0%. **Note:** If your intended answer was -2.0%, the wage growth or CPI values need adjustment.

3. **C**. Real interest rate:
$$r \approx i - \pi^e = 6.25\% - 4.0\% = 2.25\%.$$

4. **C**. Household inflation:
$$(0.48)(13) + (0.22)(8) + (0.10)(5) + (0.20)(3) = 6.24 + 1.76 + 0.50 + 0.60 = 9.10\%.$$
Closest listed answer is 8.9%.

5. **D**. Real purchasing power:
$$5\% - 9.1\% = -4.1\%.$$
Closest answer: -3%. **Note:** A closer option would be -4%.

6. **C**. Housing is the largest contributor:
$$0.48 \times 13 = 6.24 \text{ percentage points.}$$

7. **B**. Total income:
$$25,000 + 35,000 + 50,000 + 85,000 + 205,000 = 400,000.$$
Mean:
$$\frac{400,000}{5} = 80,000.$$

8. **D**. Top household share:
$$\frac{205,000}{400,000} \times 100 = 51.25\%.$$

Closest answer: 50%.

9. **C**. New total income:

$$25,000 + 35,000 + 50,000 + 85,000 + 260,000 = 455,000.$$

New mean:
$$\frac{455,000}{5} = 91,000.$$

Closest answer: $\$90,000$. **Note:** If the intended answer is $\$110,000$, the top income would need to be higher.

10. **C**. Total income remains unchanged because the transfer is financed within the sample.

11. **C**. Richest household share before redistribution:
$$\frac{205,000}{400,000} \times 100 = 51.25\%.$$

12. **B**. Bottom 40% before redistribution:

$$25,000 + 35,000 = 60,000.$$
$$\frac{60,000}{400,000} \times 100 = 15\%.$$

13. **B**. After redistribution, order incomes from lowest to highest:

$$35,000,\ 40,000,\ 50,000,\ 85,000,\ 190,000.$$

14. **C**. Richest share after redistribution:
$$\frac{190,000}{400,000} \times 100 = 47.5\%.$$

15. **A**. Before redistribution, cumulative income shares:

$$0,\ 0.0625,\ 0.1500,\ 0.2750,\ 0.4875,\ 1.0000.$$

16. **C**. Before redistribution:

$$G = 1 - \sum (Y_i + Y_{i-1})(X_i - X_{i-1}).$$

Since each household is 20%, $X_i - X_{i-1} = 0.2$.

$$\begin{aligned}\sum &= (0 + 0.0625)(0.2) + (0.0625 + 0.1500)(0.2) \\ &\quad + (0.1500 + 0.2750)(0.2) + (0.2750 + 0.4875)(0.2) \\ &\quad + (0.4875 + 1.0000)(0.2) \\ &= 0.0125 + 0.0425 + 0.0850 + 0.1525 + 0.2975 \\ &= 0.5900.\end{aligned}$$

$$G = 1 - 0.5900 = 0.4100.$$

Closest answer: 0.38. **Note:** The exact value is 0.41, so the choices should ideally include 0.41.

17. **A**. After redistribution, ordered incomes are:

$$35,000,\ 40,000,\ 50,000,\ 85,000,\ 190,000.$$

Cumulative income shares:

$$0,\ 0.0875,\ 0.1875,\ 0.3125,\ 0.5250,\ 1.0000.$$

18. **B**. After redistribution:

$$\begin{aligned}\sum &= (0 + 0.0875)(0.2) + (0.0875 + 0.1875)(0.2) \\ &\quad + (0.1875 + 0.3125)(0.2) + (0.3125 + 0.5250)(0.2) \\ &\quad + (0.5250 + 1.0000)(0.2) \\ &= 0.0175 + 0.0550 + 0.1000 + 0.1675 + 0.3050 \\ &= 0.6450.\end{aligned}$$

$$G = 1 - 0.6450 = 0.3550.$$

Closest answer: 0.30. **Note:** The exact value is 0.355, so 0.40 is also close. For a cleaner exam, include 0.36.

19. **C**. Redistribution increases the income share of lower-income households and lowers the income share of the richest household.

20. **B**. The Lorenz curve moves closer to the equality line.

21. **E**. Using:

$$G = 1 - 2A.$$

If $A = 0.42$, then:

$$G = 1 - 2(0.42) = 1 - 0.84 = 0.16.$$

22. **A**. Redistribution policies can reduce measured inequality, but their effect depends on size, targeting, and financing.

23. **C**. New high-skill wage:

$$30(1.35) = 40.50.$$

New low-skill wage:

$$20(1.06) = 21.20.$$

New wage gap:

$$40.50 - 21.20 = 19.30.$$

Closest answer: \$18.

24. **B**. This is skill-biased technological change.

25. **B**. The likely long-run labour market result is job polarization.

26. **B or C, depending on interpretation**. Gross hourly revenue is \$32, costs are \$11, so net active-hour earnings:

$$32 - 11 = 21.$$

If 20% of time is unpaid waiting time, effective earnings over total time are:

$$21(0.80) = 16.8.$$

Closest answer: \$16 or \$18. **Recommended correction:** Use answer **C** if treating closest lower rounded answer as \$16; use **D** if standard nearest rounding is intended.

27. **C**. If costs rise to \$13:

$$32 - 13 = 19.$$

Accounting for unpaid waiting time:

$$19(0.80) = 15.2.$$

Closest answer: \$16.

28. **B**. The example demonstrates monopsony power and weak worker bargaining power.

29. **D**. Price-to-income ratio:

$$\frac{1,045,000}{110,000} = 9.5.$$

30. **C**. Mortgage payment increase:

$$\frac{4,200 - 2,800}{2,800} \times 100 = \frac{1,400}{2,800} \times 100 = 50\%.$$

31. **B**. Housing affordability has worsened.

32. **D**. The mortgage rate rises from 2.3% to 6.1%. Payments would rise substantially, with the closest answer being approximately 50%.

33. **C**. If income is fixed and payments rise, the shelter-cost burden increases.

34. **B**. This shock mainly affects recent buyers and mortgage holders.

35. **C**. New rent-to-income ratio:

$$\frac{2,250}{4,500} \times 100 = 50\%.$$

36. **D**. To keep rent at 30% of income:

$$0.30Y = 2,250$$

$$Y = \frac{2,250}{0.30} = 7,500.$$

37. **B**. Higher rent reduces disposable income and household savings.

38. **B**. Lower borrowing and investment reduce aggregate demand, so GDP falls.

39. **C**. Elasticity formula:

$$\%\Delta Q_d = \varepsilon \times \%\Delta P.$$

$$-1.5 \times 10\% = -15\%.$$

Housing demand falls by 15%.

40. **B**. Exchange rate appreciation lowers import prices and reduces inflationary pressure.

41. **B**. Total production cost increase:

$$25\% \times 20\% = 5\%.$$

42. **B**. This is cost-push inflation.

43. **B**. Higher production costs reduce output.

44. **B**. Net demand pressure:

$$3\% - 1\% = 2\%.$$

45. **B**. With low supply elasticity, prices rise sharply.

46. **B**. The likely effect is higher housing-related inequality.

47. **C**. Housing demand elasticity:

$$-1.2 \times 3\% = -3.6\%.$$

Closest answer: 4%.

48. **B**. If demand falls while supply is fixed, house prices likely fall.

49. **B**. Rental demand likely rises as some households are priced out of ownership.

50. **A**. Real wage growth:

$$6\% - 5\% = 1\%.$$

51. **B**. If inflation persists, wage demands are likely to rise.

52. **B**. The likely macroeconomic outcome is persistent inflation.

53. **C**. If home values double, the increase is 100%.

54. **B**. Renters' relative wealth position worsens.

55. **B**. Overall wealth inequality rises.

56. **B**. More housing supply puts downward pressure on house prices.

57. **B**. More rental supply puts downward pressure on rents.

58. **B**. Housing-related inequality likely falls.

59. **B**. Higher firm costs reduce profits, all else equal.

60. **C**. The employment effect is ambiguous because higher costs may reduce hiring, but better wages may improve retention and productivity.

61. **A**. Worker welfare likely rises because effective wages and protections improve.

62. **B**. During a recession, weaker labour demand tends to reduce wage growth or wages.

63. **B**. Consumption likely falls.

64. **B**. Inequality may rise because lower-income and precarious workers are often hit hardest.

65. **A**. This is substitution bias.

66. **A**. The official CPI overstates inflation because it does not fully capture substitution toward cheaper goods.

67. **B**. Higher Canadian rates attract capital inflows.

68. **B**. Increased demand for Canadian assets appreciates the Canadian dollar.

69. **A**. A stronger Canadian dollar makes imports cheaper.

70. **B**. Higher real interest rates reduce investment.

71. **B**. Lower investment and borrowing reduce GDP.

72. **B**. Borrowing falls.

73. **C**. Real wage growth:
$$4\% - 6\% = -2\%.$$

74. **B**. Housing affordability worsens.

75. **B**. The overall outcome is volatility and policy trade-offs.